

CollaDrum: Designing a Tangible Interactive Sonic System to Foster Collaboration in Children with Autism Spectrum Disorder

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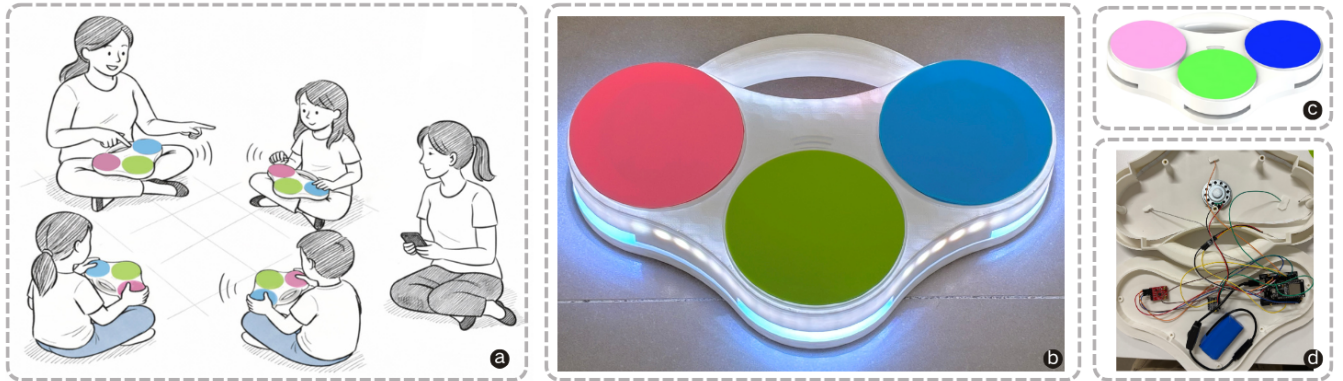


Figure 1: (a) a sketch classroom scenario using *CollaDrum* for enhancing task-based collaboration. (b) The prototype of *CollaDrum* with visual feedback. (c) A 3D enclosure rendering of *CollaDrum*. (d) An internal view of *CollaDrum* showing the hardware design.

Abstract

Children with Autism Spectrum Disorder (ASD) often experience difficulties in collaborative classroom activities. However, few technologies build upon interaction modalities and collaborative game rules that children are already accustomed to in classroom contexts to promote meaningful peer collaboration. To address this gap, we conducted interviews with five educators and, informed by their classroom experiences, co-designed a tangible interactive system, *CollaDrum*. The system integrates task-based collaborative games

embedded within individual tangible devices, together with a mobile interface that enables educators to coordinate and intervene during activities. We further conducted an exploratory pilot study with both high-functioning and low-functioning children with ASD to obtain preliminary insights into the system's influence on collaborative engagement and to identify design issues for iterative refinement. Finally, we reflect on the design process and pilot study, discussing how task-based tangible collaboration may support more socially meaningful interaction for children with ASD.

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CCS Concepts

• Human-centered computing → Accessibility systems and tools; Interaction devices.

Keywords

Autism spectrum disorder; collaboration; tangible interfaces; classroom technologies.

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1 Introduction

Collaboration is widely recognized as an essential development target for children with Autism Spectrum (ASD) [4–6]. In classroom settings, collaboration is often scaffolded through structured, task-based activities that promote joint attention and social interaction [5, 12, 15]. To support such activities, children frequently engage with tangible sensory tools in collaborative classroom activities, such as drums and other musical instruments used in music sessions, participating through turn-taking and other structured interaction rules [7]. Through repeated participation in these scaffolded interactions, children gradually develop the capacity for more spontaneous forms of peer coordination [2].

In recent years, researchers have explored technological approaches to augment collaborative engagement for children with ASD. Early foundational work established the viability of shared digital spaces, demonstrating how tabletop systems can effectively structure social behaviors [5, 6]. Building on these principles, subsequent research has shifted toward tangible and multimodal interfaces. Beyond musical contexts, interactive systems such as iTILES [10], RainbowForest [13] and recent classroom-adapted tangibles like Co-Space [16] embed social mechanics, including role assignment, turn-taking, and resource negotiation, to encourage collaborative behaviors during play. Together, these works demonstrate how tangible and multimodal interaction can be designed to structure and support collaboration. However, many of these systems adopt interaction paradigms or rule structures that differ substantially from those familiar to children in classroom settings. Mastering these unfamiliar interfaces can increase the operational and cognitive burden, inadvertently shifting children's attention toward understanding the device rather than engaging with their peers, which may ultimately undermine natural collaborative dynamics.

Informed by educator interviews, we designed *CollaDrum*, a tangible interactive sonic system that aligns with familiar classroom interaction patterns while embedding collaborative scaffolding within individual multimodal devices. An exploratory pilot study with children with ASD revealed both opportunities for spontaneous coordination and challenges related to sensory demands and feedback intensity, which further shaped our design. This work contributes to collaborative technologies for children with ASD by demonstrating how embedding collaborative scaffolding within individual, multimodal devices can foster meaningful social interaction while reducing reliance on direct educator intervention.

2 Formative Study

To design a tangible interactive device aligned with children's established classroom practices and supportive of collaborative engagement among children with ASD, we conducted a formative study grounded in a human-centered design approach. We carried

out semi-structured interviews with educators to identify design opportunities grounded in their existing classroom practices for organizing and facilitating collaborative activities. The study was approved by the authors' university Institutional Review Board (ID: FWA00021580) and conducted in collaboration with a local private educational institution serving children with ASD. The partner institution supported study coordination, including scheduling, communication with educators and parents, and organization of classroom activities.

2.1 Interviews

With the assistance of the institution's coordinator, we recruited five educators (4 female, 1 male) with 1–10 years of teaching experience from the partner institution. All participants had direct experience supporting collaborative classroom activities for children with ASD. For reporting purposes, we refer to the educators as T1–T5 throughout the paper. We conducted semi-structured interviews with these educators across three sessions, including two small-group interviews (T1–T2; T4–T5) and one individual interview (T3). The interviews lasted 45–60 minutes and were structured to support design-oriented reflection on everyday classroom practices, helping surface limitations in existing collaborative activities and explore opportunities for tangible-supported collaboration. This process informed the identification of a set of design considerations.

2.2 Data Collection and Analysis

All semi-structured interviews were audio-recorded with participants' consent and securely stored on an internal cloud platform to ensure data privacy. The recordings were subsequently transcribed verbatim for analysis. We analyzed the interview data using an inductive thematic analysis approach [3], supported by ATLAS.ti for organizing and coding qualitative data. Two authors independently conducted open coding on 30% of the interview transcripts, identifying preliminary themes. They then met to compare and discuss their codes, resolve discrepancies, and refine the coding scheme, resulting in the development of a shared codebook. Using the finalized codebook, the first author coded the remaining transcripts.

2.3 Findings

Through semi-structured interviews with the educators, we derived several design considerations (*DCs*) to develop a tangible interactive device to support collaborative activities among children with ASD in classroom contexts.

DC1. Attention Guidance and Regulation during Collaboration. Educators emphasized that children with ASD often struggle with distractibility or overfocus during collaborative activities (T1–T5), highlighting the need for attention support. *First*, sensory cues should be carefully calibrated to sustain interest without becoming distracting, as overly complex visuals or intense lighting can divert attention from collaboration [8, 14]. *Second*, educators noted that device-mediated attention regulation is less interruptive and more self-initiated than verbal reminders, supporting sustained engagement (T4, T5).

DC2. Encouraging Active Awareness of Peer States. Educators emphasized the need to encourage children to attend to their

peers rather than relying solely on external prompts (T2, T5). While system-driven cues can guide individual actions, they often trap children in isolated, parallel play rather than fostering genuine shared collaboration [6, 11]; educators therefore closely observed whether children oriented their attention toward peers during interaction. Based on these insights, educators highlighted the importance of interaction rules that require awareness of others' states beyond system-provided guidance.

DC3. Increasing Challenge to Support Engagement. Educators emphasized that incorporating multiple levels of difficulty enables children with different functional abilities to participate meaningfully in collaborative activities (T1, T2, T4, T5). Simplified rules and supportive cues are essential for children with lower functional levels, while progressively increasing task difficulty helps sustain engagement and motivation for higher-functioning children (T5). Educators further noted that gradual challenge supports repeated practice and reinforces understanding of collaborative rules [9].

DC4. Designing Explicit Feedback for Task Success and Failure. Educators emphasized that in structured, task-based activities, collaboration should be accompanied by explicit success and failure signals (T1–T5). They noted that introducing perceptible signals to indicate successful task completion or breakdowns in coordination enables children to recognize and respond to collaborative states. Success signals reinforce coordinated participation and sustain engagement, whereas failure signals indicate that shared task requirements have not yet been met, prompting behavioral adjustment. Rather than serving merely as reward or punishment, these signals function as regulatory mechanisms that guide coordination and support adaptive collaboration [4].

3 System Design

Informed by educator interviews and sustained collaboration with an experienced special educator (8+ years), we designed an integrated system that combines distributed tangible sonic devices with a mobile educator interface to support collaborative engagement among children with ASD, as illustrated in Figure 1–a. The educator's continuous involvement enabled iterative refinement of the interaction framework, ensuring alignment with pedagogical goals and technical feasibility, while grounding the collaboration structure in game formats already familiar within children's classroom routines.

Our system design comprises four interconnected components. (1) *Collaboration-Oriented Design Grounding*: The system is grounded in educators' descriptions of everyday classroom collaboration practices and interview-derived design considerations, aligning both individual device interaction and multi-device coordination with interaction forms familiar to children. (2) *Child-Device Interaction*: Inspired by classroom use of percussion instruments, we designed distributed, movable tangible devices that support pad tapping and shaking-based interaction, with multiple sound timbres to capture attention and sustain engagement (**DC1**). (3) *Device-Mediated Collaboration between Children*: Based on collaborative game formats that children regularly engage in during classroom sessions, we implemented collaborative game rules centered on turn-taking and simultaneous play, guided by colored light cues and multimodal

sound–light feedback to signal task success or failure, reduce interaction difficulty and reinforce coordination (**DC1**, **DC3**, **DC4**). Activities are conducted in small groups of three children with an educator acting as a guiding collaborator. (4) *Educator-Side Facilitation*: A mobile educator interface enables a second educator to monitor interactions and provide timely support, including switching activities, adjusting difficulty, muting devices, and triggering reminder cues to guide attention (**DC1**, **DC3**).

3.1 Hardware Design

We designed the system's tangible device around an ESP32 microcontroller, supporting bodily input, multi-sensory output, and wireless communication for collaborative interaction, as shown in Figure 1–d.

Physical Form. The device features a cloud-shaped, snap-fit enclosure with rounded edges to ensure safe classroom use. The devices were first designed through 3D modeling (Figure 1–c) and then fabricated via 3D printing in PLA. Identical devices are used by both children and educators. Each device includes three raised silicone-covered pads that provide soft tactile feedback for tapping, as well as a dedicated handle for grasping and shaking interactions; different pads and shaking gestures trigger distinct sound timbres. To reduce visual distraction, the enclosure is neutral white, while the pads are colored blue, green, and red to provide clear visual cues with sufficient salience (**DC1**) [14].

Input & Output Design. Tapping input is captured using the ESP32's capacitive touch interface, while shaking is detected via an onboard MPU6050 inertial measurement unit. The drum-inspired tapping and shaking interactions leverage children's existing familiarity with classroom percussion instruments, eliminating the need to learn complex new operation modes, minimizing operational burden, and allowing children to focus on peer collaboration rather than system mastery. Visual feedback is delivered through an internal WS2812 RGB LED strip, producing softly diffused, non-intrusive illumination to guide attention without overstimulation (**DC1**), as presented in Figure 1–b. Auditory feedback is provided by a DFPlayer Mini module, enabling both musical and imaginative non-instrumental sounds (e.g., animal effects) to support multisensory differentiation (**DC1**).

3.2 Software Design

The software layer coordinates interaction across multiple tangible devices and the educator-side interface to support collaborative classroom activities.

Device Connectivity. The system consists of four tangible devices, a mobile educator-side interface, and a central coordination computer. All components operate within the same local wireless network. Each tangible device captures interaction inputs through its ESP32 microcontroller and transmits them via Wi-Fi to the coordination computer. The coordination computer maintains shared interaction states and processes game logic before returning corresponding audio and visual feedback to the devices, enabling synchronized and coordinated collaborative interaction.

Educator-Side Interface Design. To support classroom facilitation and real-time intervention, we designed a mobile educator interface that synchronizes with all tangible devices over a local Wi-Fi network. The interface enables educators to coordinate collaborative activities by switching game modes and sound sets, muting devices, or triggering synchronized light–sound reminders to guide attention and mitigate overstimulation (**DC1**). It also allows dynamic adjustment of difficulty parameters, such as reaction-time constraints, enabling activities to adapt to children’s real-time progress and collaborative performance (**DC3**).

3.3 Game Modes Design

Drawing on collaborative behaviors of turn-taking and simultaneous play observed in classroom activities and prior research, we co-design two game modes with educators to integrate into the tangible device system. Both modes feature clearly defined goals and explicit rules to structure collaborative interaction, and incorporate feedback signals indicating task success or failure to guide coordination (**DC4**). To accommodate the heterogeneous functional abilities of children with ASD, including both high-functioning and low-functioning groups, the game modes incorporate progressive difficulty adjustments (**DC3**). Completion of each activity requires children to actively attend to and observe their peers, thereby fostering collaborative engagement throughout gameplay (**DC2**). All game modes are initiated and configured through the educator’s mobile-based interface.

Turn-Taking Game. This game mode is proposed by educators in interviews and used in prior work [2]. **Goal:** The group collaboratively reproduces a sequence of sound triggers in turn. **Rules:** At each round, one child is active (indicated by a white light) while others observe; subsequent participants reproduce the sequence using their own devices, promoting attention to peers and memory of shared interactions. **Feedback:** Successful reproduction by all participants trigger celebratory sounds and green lights, while out-of-turn actions, incorrect sounds, or prolonged inactivity result in red light cues and reset the sequence. **Difficulty:** The game includes five difficulty levels, increasing sequence length per round, and educators may optionally adjust timing constraints to match group performance.

Simultaneous-Play Game. Drawing on prior work on joint attention in collaborative play [1], we proposed this mode co-designed with educators to naturally integrate into classroom activities. **Goal:** The game requires designated children to activate their devices within the same temporal window, promoting attention to peers’ actions and synchronous coordination. **Rules:** At each round, a white light scrolls across devices and stops at one or more participants (green light), who must interact simultaneously using any valid input (e.g., pad tapping or shaking), while others remain inactive; a round succeeds only if all required interactions occur within the allowed time window. **Feedback:** Success depends on temporal alignment, triggering celebratory sounds and green lights whereas out-of-time or out-of-turn actions result in red lights and a round restart. **Difficulty:** Difficulty spans four levels, increasing both the number of participants (1–4) and the temporal precision required,

with higher levels demanding tighter coordination within shorter time windows.

4 Pilot study and Reflections

We conducted an exploratory, formative pilot study with two groups of children with ASD, including a low functioning group ($N = 3$) and a high functioning group ($N = 3$). The goal was to gain early insights into how the tangible system influenced collaborative engagement in classroom activities and to identify design and functional issues that could inform subsequent iterations. *First*, the system showed differentiated effects on collaborative engagement between the two groups. The low functioning group experienced difficulties in rule comprehension and showed limited responsiveness to sensory feedback, whereas the high functioning group demonstrated rapid understanding of device use and more active collaborative participation. Notably, children in the high functioning group attended to peers’ interactions, verbally expressed collaborative intentions, and showed increased motivation to attempt higher difficulty levels. *Second*, several functional limitations were identified that require further iteration. Some children exhibited anxiety or avoidance toward failure-related feedback, particularly when unsuccessful attempts were indicated by red lights and buzzer sounds. Educators suggested replacing these with gentler, more encouraging sensory cues to reduce frustration.

Drawing on the design process and feedback from the pilot study, we present the following reflections.

Adapting interaction design to children’s existing habits. Rather than introducing unfamiliar interaction paradigms, our design was informed by how children with ASD already engage with tangible materials during everyday social activities. For example, the tapping and shaking interactions of the device were derived from children’s habitual use of traditional musical instruments. Findings from our pilot study suggest that integrating such familiar interaction techniques and interaction logic enabled children to more quickly understand how to use the system, thereby reducing cognitive demands. This, in turn, allowed children to allocate more attention to collaborative engagement rather than to learning new interaction rules.

Supporting collaboration for both children and educators. In our design, we considered educators’ needs alongside children’s participation. The system supported different educator roles, allowing one educator to participate using the same tangible device as the children, while another coordinated the activity through the mobile interface. This configuration facilitated collaboration not only among children but also between children and educators. Importantly, the mobile interface enabled guidance to be embedded within the activity rather than delivered through explicit instruction, reducing children’s perception of teacher imposed demands. This suggests that collaborative systems should support educators’ practical needs in addition to enabling children’s participation.

5 Conclusion

In this paper, we introduced *CollaDrum*, a tangible interactive sonic system co-designed with educators to support task-based, mutually responsive collaboration among children with ASD in classroom

settings. Through an exploratory pilot study, we showed how embedding collaborative scaffolding within distributed, multimodal devices can elicit joint attention and spontaneous verbal coordination while reducing reliance on direct educator intervention. At the same time, we emphasize that these findings are preliminary. This work highlights critical challenges regarding sensory feedback design and the necessity of adapting to diverse ASD subtypes. These insights lay the groundwork for rigorous, longitudinal case studies in the future, which will quantitatively evaluate the system's impact on social initiative and further refine technologies that support inclusive classroom ecology.

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